INS 3121
SOIL MECHANICS

Shear Strength of Soils

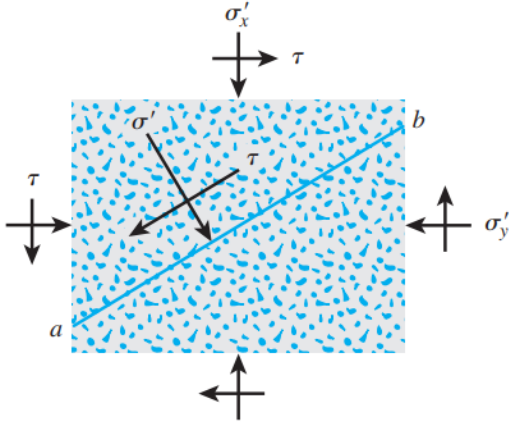
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Introduction

- The shear strength of a soil mass is the internal resistance per unit area that the soil mass can offer to resist failure and sliding along any plane inside it.
- In order to analyze soil stability problems, such as bearing capacity, slope stability, and lateral pressure on earth-retaining structures



Mohr–Coulomb Failure Criterion

- A material fails because of a critical combination of normal stress and shearing stress and not from either maximum normal or shear stress alone (Mohr, 1900).

- The failure envelope is a curved line.
- For most soil mechanics problems, it is sufficient to approximate the shear stress on the failure plane as a linear function of the normal stress (Coulomb, 1776).

(Mohr–Coulomb failure criterion)

Where c = cohesion

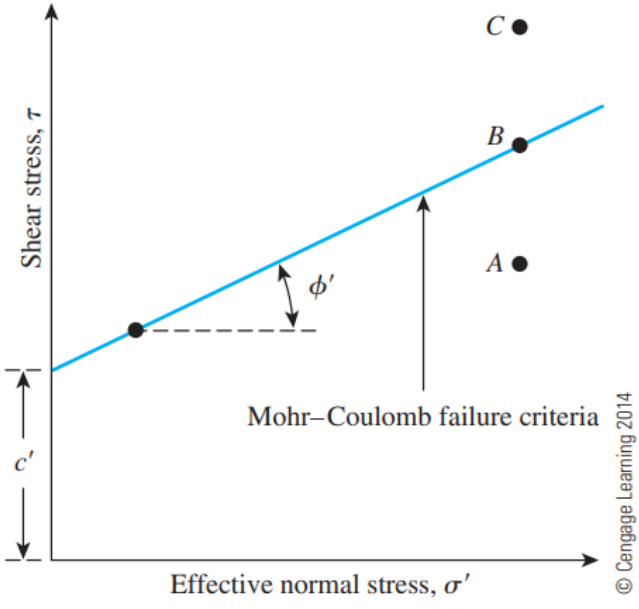
= angle of internal friction

= normal stress on the failure plane

= shear strength

Mohr–Coulomb Failure Criterion

- In saturated soil, the total normal stress at a point is the sum of the effective stress () and pore water pressure (u),
- The effective stress is carried by the soil solids.
- Mohr–Coulomb failure criterion, expressed in terms of effective stress
- where cohesion and friction angle, based on effective stress.



Mohr–Coulomb Failure Criterion

- Mohr–Coulomb failure criterion, expressed in terms of effective stress



- The value of for sand and inorganic silt is 0.
- For normally consolidated clays, can be approximated at 0.
- Overconsolidated clays have values of that are greater than 0.
- The angle of friction, , is sometimes referred to as the drained angle of friction.

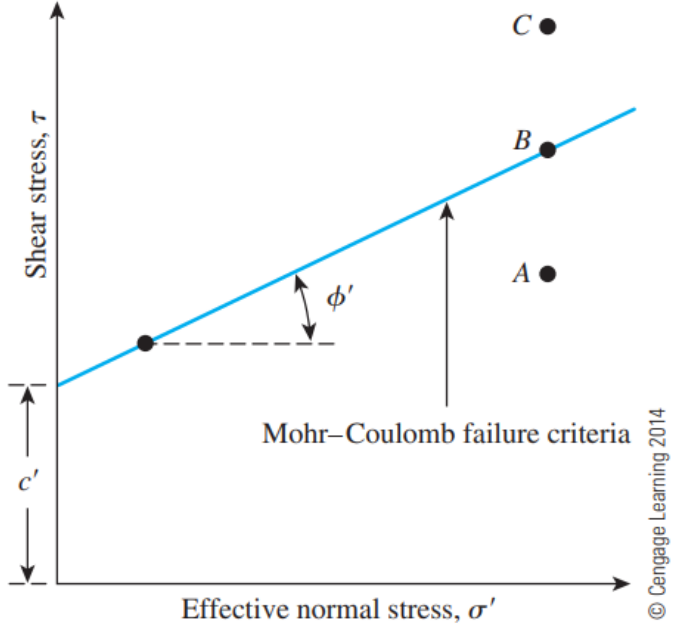
Mohr–Coulomb Failure Criterion

Table 12.1 Typical Values of Drained Angle of Friction for Sands and Silts

Soil type	ϕ' (deg)
Sand: Rounded grains	
Loose	27–30
Medium	30–35
Dense	35–38
Sand: Angular grains	
Loose	30–35
Medium	35–40
Dense	40–45
Gravel with some sand	34–48
Silts	26–35

Mohr–Coulomb Failure Criterion

- Point A, shear failure will not occur along the plane.
- Point B (which falls on the failure envelope), shear failure will occur along that plane
- Point C cannot exist, because it plots above the failure envelope, and shear failure in a soil would have occurred already.



- The failure plane *EF* makes an angle with the major principal plane.
- *fgh* is the failure envelope defined by the relationship
- The radial line *ab* defines the major principal plane (CD in Figure 12.2)
- The radial line *ad* defines the failure plane (EF in Figure 12.2)

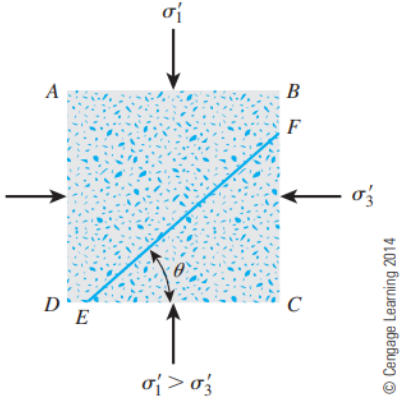


Figure 12.2 Inclination of failure plane in soil with major principal plane

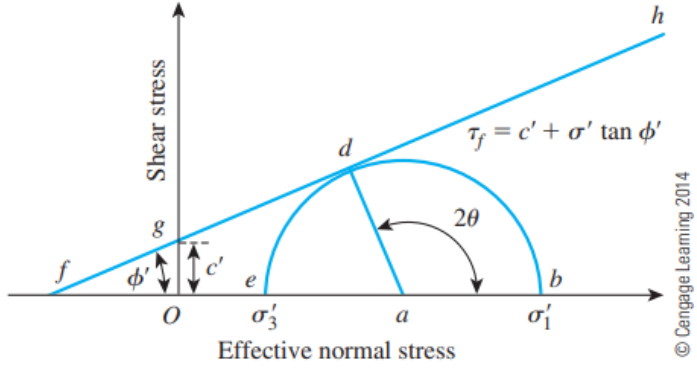


Figure 12.3 Mohr's circle and failure envelope

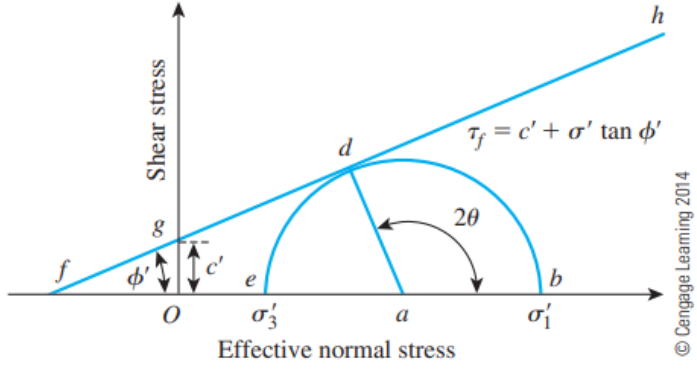


Figure 12.3 Mohr's circle and failure envelope

or,

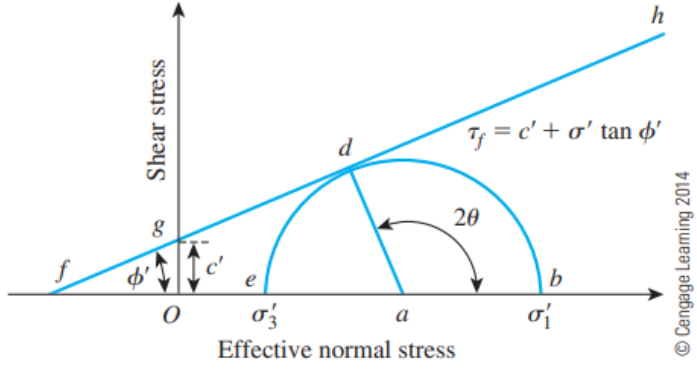


Figure 12.3 Mohr's circle and failure envelope

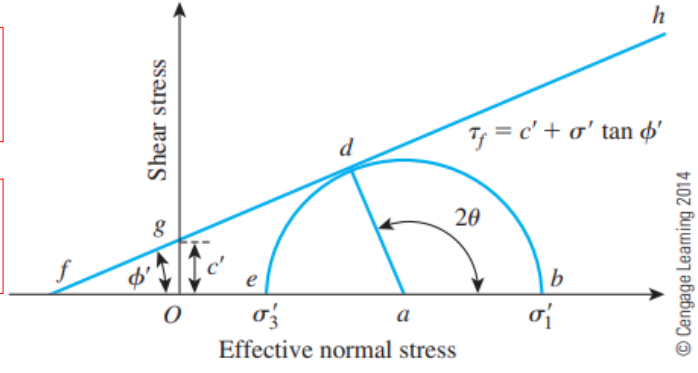


Figure 12.3 Mohr's circle and failure envelope

Laboratory Test for Determination of Shear Strength Parameters

- Several methods to determine the shear strength parameters() of various soil specimens in the laboratory
  - Direct shear test
  - Triaxial test
  - Direct simple shear test
  - Plane strain triaxial test
  - Torsional ring shear test

The direct shear test and triaxial test are the two commonly used techniques for determining the shear strength parameters.

Direct Shear Test

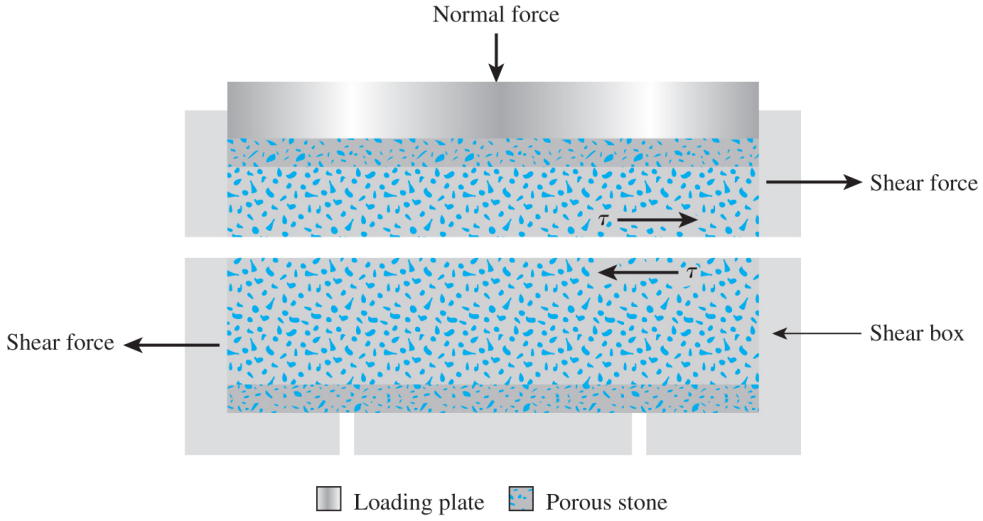


Figure 12.4 Diagram of direct shear test arrangement

Direct Shear Test

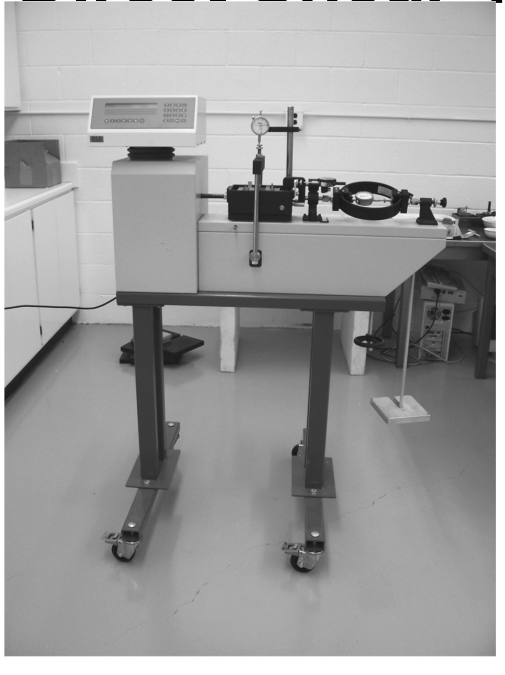


Figure 12.5 Strain-controlled direct shear equipment (*Courtesy of Braja M. Das, Henderson, Nevada*)

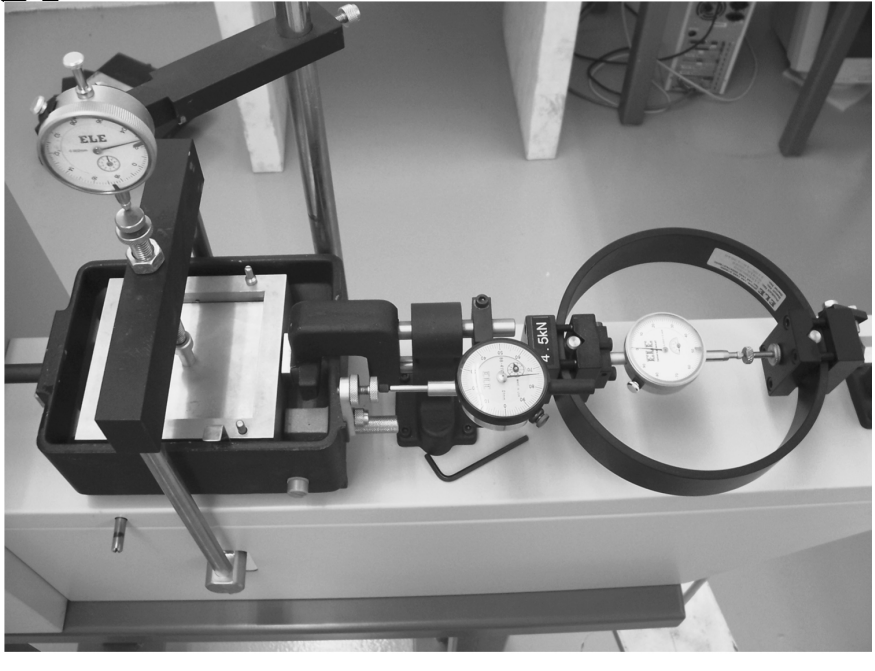


Figure 12.6 A photograph showing the dial gauges and proving ring in place (*Courtesy of Braja M. Das, Henderson, Nevada*)

Direct Shear Test

- The soil specimens may be square or circular in plan. The size of the specimens generally used is about 51mmX51mm or 102mm X102mm across and about 25mm high.
- The box is split horizontally into halves.
- Stress – controlled or strain controlled test.

- The advantage of strain – controlled tests is that, in the case of dense sand, peak shear resistance (that is, at failure) as well as lesser shear resistance (that is, at a point after failure called *ultimate strength*) can be observed and plotted.

Direct Shear Test

- In stress – controlled tests, only the peak shear resistance can be observed and plotted.
- Peak shear resistance in stress – controlled tests can be only approximated because failure occurs at a stress level somewhere between the pre-failure load increment and the failure load increment.

σ = Normal stress =

τ = Shear stress =

Direct Shear Test

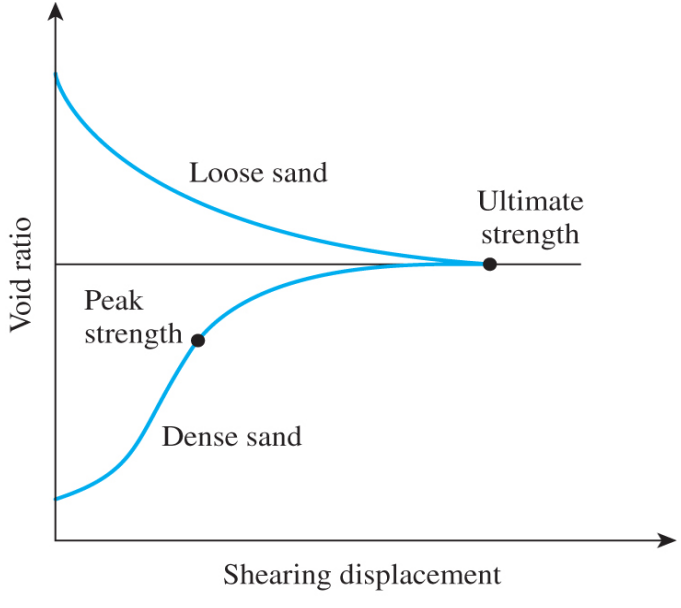


Figure 12.8 Nature of variation of void ratio with shearing displacement

Direct Shear Test

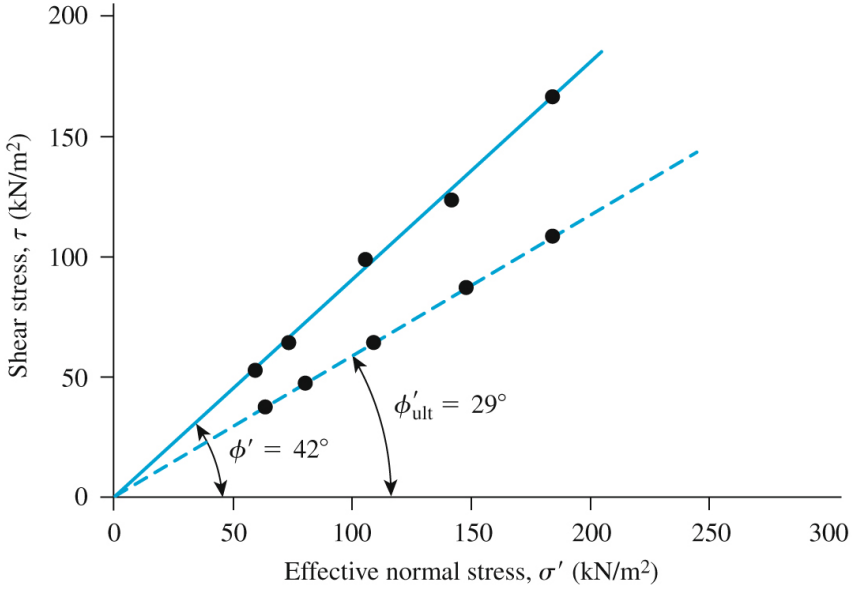


Figure 12.9 Determination of shear strength parameters for a dry sand using the results of direct shear tests

Direct Shear Test

- The following generalization can be developed from the figure

1. In loose sand, the resisting shear stress increases with shear displacement until a failure shear stress of τ_f is reached.

   After that, the shear resistance remain approximately any further increase in the shear displacement.

2. In dense sand, the resisting shear stress increase with shear displacement until it reaches a failure stress of τ_f. This τ_f is called the peak shear strength.

   After failure stress is attained, the resisting shear stress gradually decreases as shear displacement increase until it finally reaches a constant value called the ultimate shear strength.

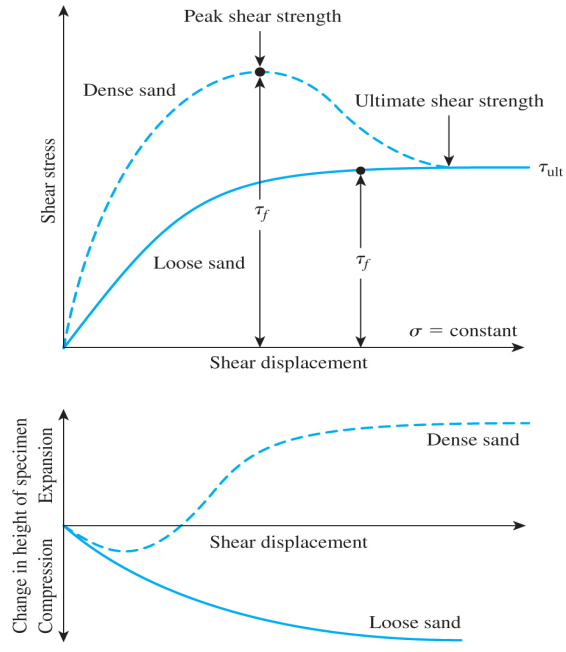
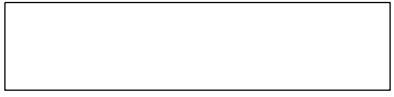
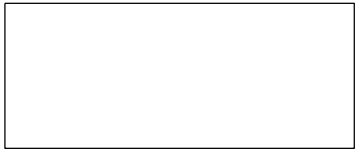


Figure 12.7 Plot of shear stress and change in height of specimen against shear displacement for loose and dense dry sand (direct shear test)

Direct Shear Test

- Test in a dry sand





It is important to note that *in situ* cemented sands may show a ***c'*** intercept

Direct Shear Test

- The ultimate shear strength ()
-

Drained Direct Shear Test on Saturated Sand and Clay

- The ultimate shear strength ()
-

Drained Direct Shear Test on Saturated Sand and Clay

- A drained test is made on a saturated soil specimen by keeping the rate of loading slow enough so that the excess pore water pressure generated in the soil is completely dissipated by drainage.
- The friction angle ϕ' obtained from a drained direct shear test of saturated sand will be the same as that for a similar specimen of dry sand.

Drained Direct Shear Test on Saturated Sand and Clay

- The test may last from 2 to 5 days.
- Note that $\sigma = \sigma'$ and the value of $c' \approx 0$ for a normally consolidated clay.

Drained Direct Shear Test on Saturated Sand and Clay

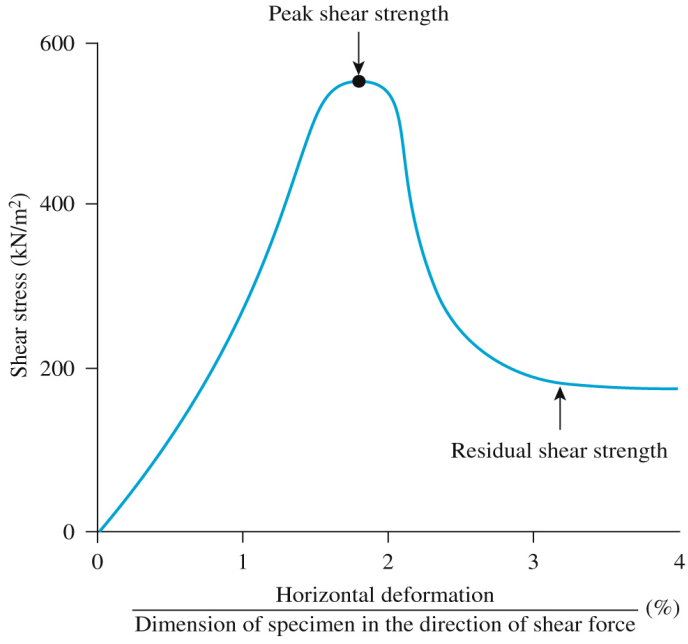


Figure 12.10 Results of a drained direct shear test on an overconsolidated clay [*Note:* Residual shear strength in clay is similar to ultimate shear strength in sand (see Figure 12.7)]

Drained Direct Shear Test on Saturated Sand and Clay

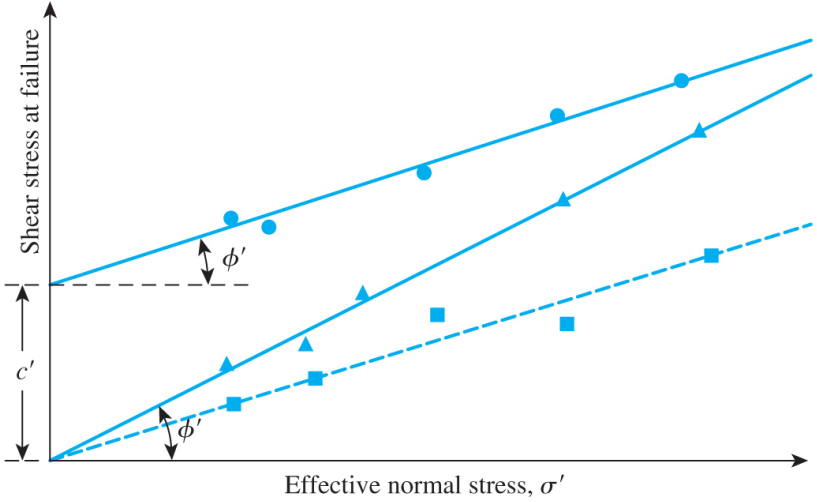


Figure 12.11
Failure envelope for clay obtained from drained direct shear tests

Drained Direct Shear Test on Saturated Sand and Clay

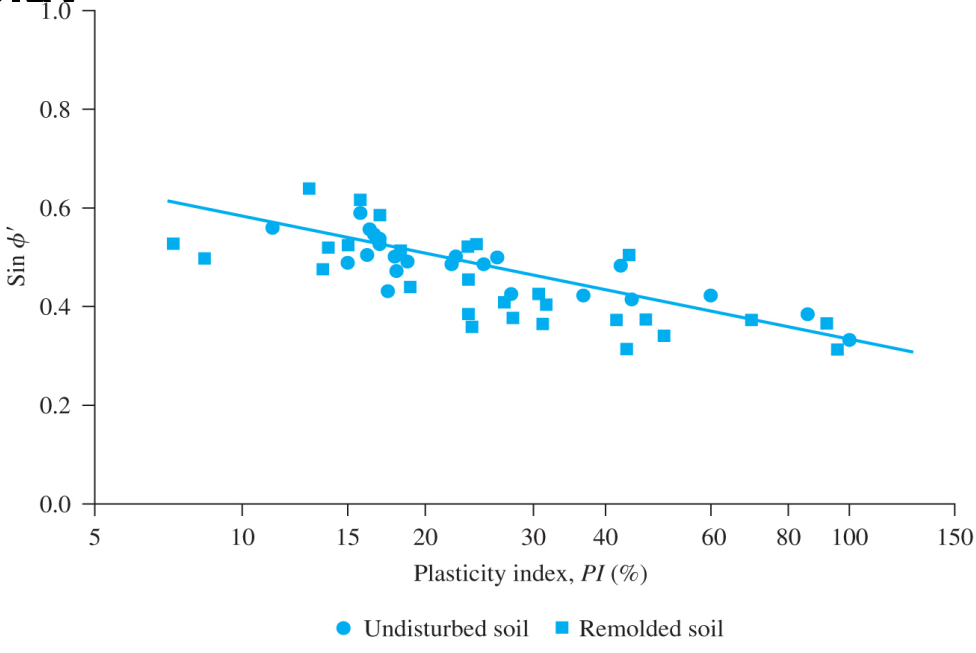


Figure 12.12 Variation of sin ϕ' with plasticity index for a number of soils (*After Kenney, 1959. With permission from ASCE.*)

General Comments on Direct Shear Test

- The direct shear test is rather simple to perform, but it has some inherent shortcomings.
  1. Soil is not allowed to fail along the weakest plane but is forced to fail along the plane of split of the shear box.
  2. The shear stress distribution over the shear surface of the specimen is not uniform.

In many foundation design problems, it will be necessary to determine the angle of friction between the soil and the material in which the foundation is constructed.

Triaxial Shear Test - General

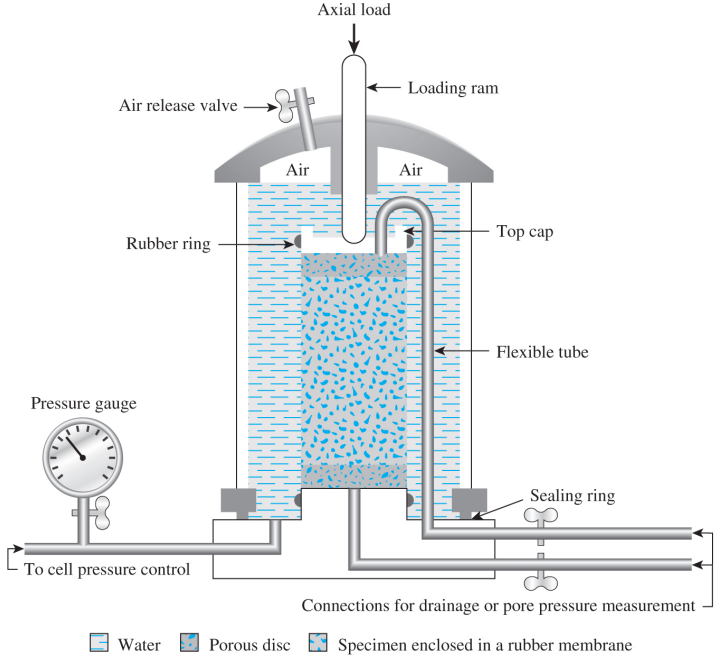


Figure 12.19 Diagram of triaxial test equipment (*After Bishop and Bjerrum, 1960. With permission from ASCE)*

Triaxial Shear Test - General

- In this test, a soil specimen about 35.6 mm in diameter and 76.2mm in length is generally used.
- confining pressure
- axial stress (sometimes called *deviator stress)*

Triaxial Shear Test - General

- This can be done in one of two ways:
  1. Application of dead weights or hydraulic pressure in equal increments until the specimen fails.– Stress controlled test

  2. Application of axial deformation at a constant rate by means of a geared or hydraulic loading press.
     This a strain – controlled test

- Three standard types of triaxial tests :
  3. Consolidated – drained test or drained test(CD test)
  4. Consolidated – undrained test (CU test)
  5. Unconsolidated – undrained test or undrained test (UU test)

Consolidated – Drained Triaxial Test

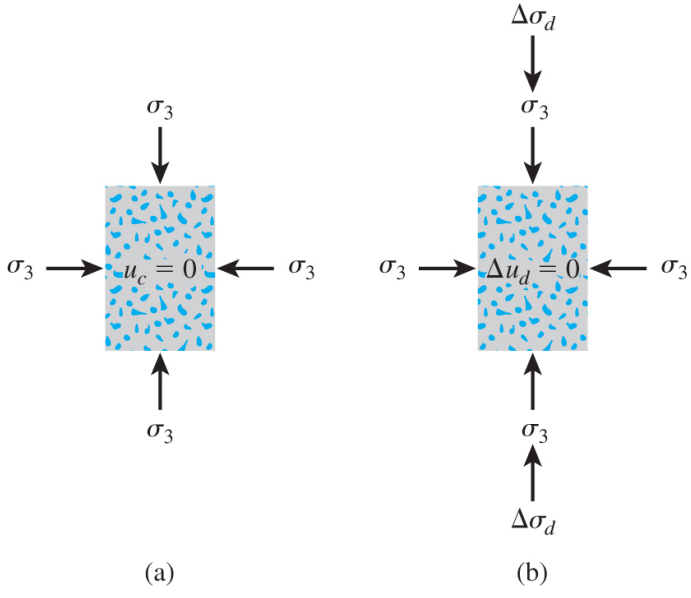


Figure 12.20
Consolidated-drained triaxial test: (a) specimen under chamber-confining pressure; (b) deviator stress application

Consolidated – Drained Triaxial Test

- Confining pressure σ_3

$$B = \frac{u_c}{\sigma_3}$$

where B = Skempton's pore pressure parameter

(Skempton, 1954)

- For saturated soft soils, B is approximately equal to 1

Consolidated – Drained Triaxial Test

Table 12.2 Theoretical Values of B at Complete Saturation

Type of soil	Theoretical value
Normally consolidated soft clay	0.9998
Lightly overconsolidated soft clays and silts	0.9988
Overconsolidated stiff clays and sands	0.9877
Very dense sands and very stiff clays at high confining pressures	0.9130

- If drainage is opened, with time, will become equal to 0.

In saturated soil, the change in the volume of the Specimen (ΔVc) that takes place during consolidation can be obtained from the volume of pore water drained

Consolidated – Drained Triaxial Test

Table 12.2 Theoretical Values of B at Complete Saturation

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Consolidated – Drained Triaxial Test

- If drainage is opened, with time, will become equal to 0.

In saturated soil, the change in the volume of the Specimen (ΔVc) that takes place during consolidation can be obtained from the volume of pore water drained (Figure 12.21a)

- Next, the deviator stress, $\Delta\sigma_d$, on the specimen is increased at a very slowly ($\Delta u_d = 0$)
- A typical plot of the variation of deviator stress against strain in loose sand and normally consolidated clay is shown in Figure 12.21b

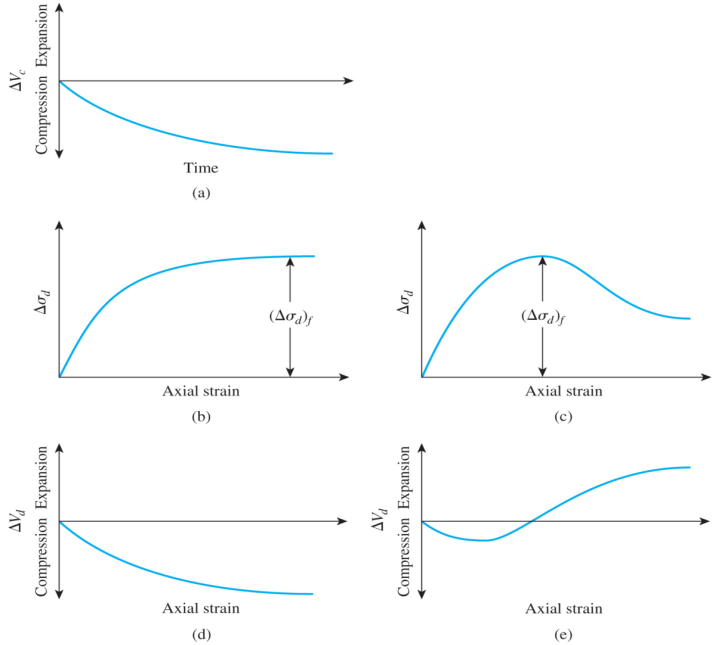


Figure 12.21 Consolidated-drained triaxial test: (a) volume change of specimen caused by chamber-confining pressure; (b) plot of deviator stress against strain in the vertical direction for loose sand and normally consolidated clay; (c) plot of deviator stress against strain in the vertical direction for dense sand and overconsolidated clay; (d) volume change in loose sand and normally consolidated clay during deviator stress application; (e) volume change in dense sand and overconsolidated clay during deviator stress application

Consolidated – Drained Triaxial Test

- Figure 12.21c show a similar plot for dense sand and overconsolidated clay
- The volume change, ΔV_d , of specimens that occurs because of the application of deviator stress in various soils is also shown in Figure 12.21d and e.
- Fig 12.21 (d) volume change in loose sand and normally consolidated clay during deviator stress application;
- Fig.12.22 (e) volume change in dense sand and overconsolidated clay during deviator stress application

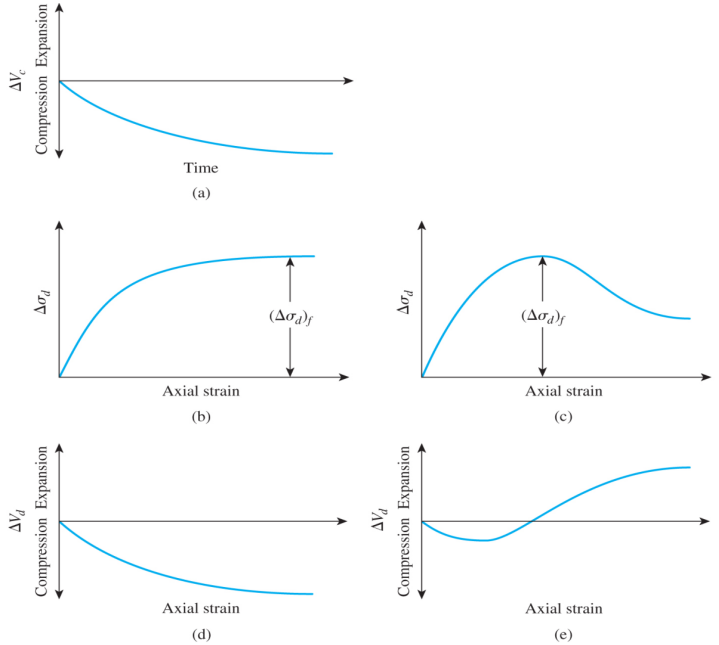


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Consolidated – Drained Triaxial Test

- total and effective confining stress =$\sigma_3 = \sigma_3'$
- total and effective axial stress at failure σ_3 $(\Delta\sigma_d)_f$ σ_1 σ_1'

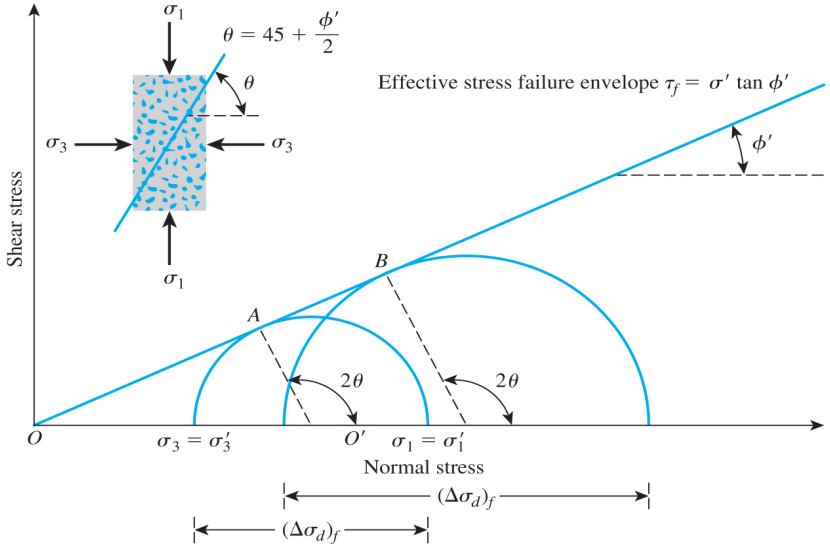


Figure 12.22 Effective stress failure envelope from drained tests on sand and normally consolidated clay

Consolidated – Drained Triaxial Test

- Overconsolidation results when a clay is initially consolidated under an all – around chamber pressure of σ_c ($=\sigma_c'$) and is allowed to swell by reducing the chamber pressure to σ_3 ($=\sigma_3'$).
- OC clay shows 2 distinct branches of failure envelope
- A consolidated-drained triaxial test on a clayey soil may take several days to complete
- For that reason, the CD type of triaxial test is not common.

Consolidated – Drained Triaxial Test

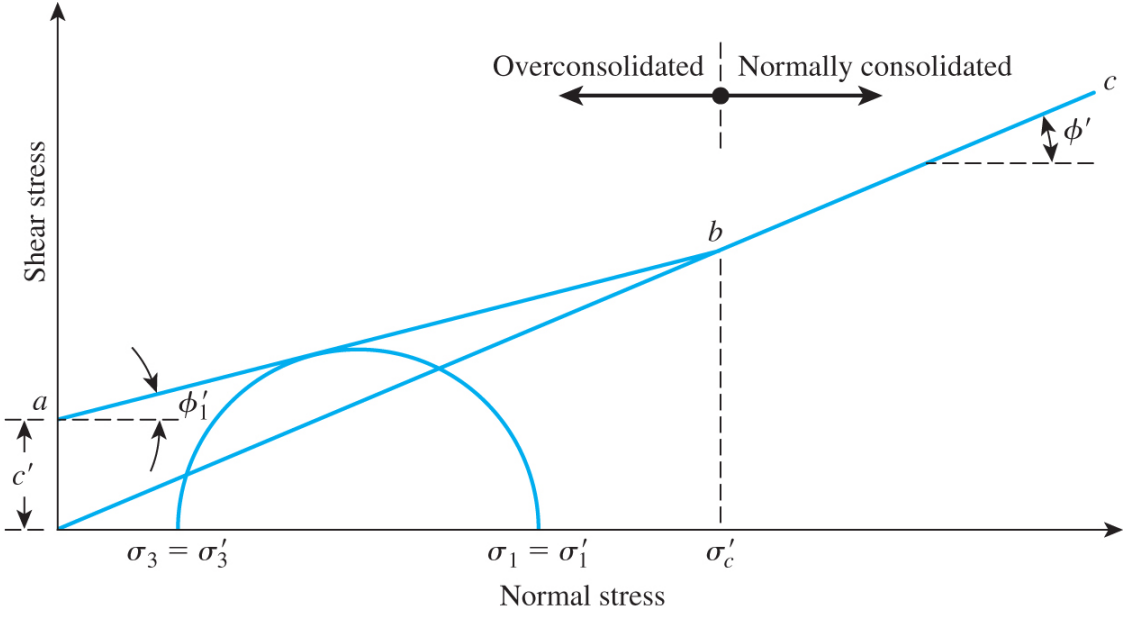


Figure 12.23 Effective stress failure envelope for overconsolidated clay

Consolidated – Undrained Triaxial Test

- CU test is the most common type of triaxial test.

- In this test, the saturated soil specimen is first consolidated by all-round chamber fluid pressure.

- After the pore water pressure generated by the application of confining pressure is completely dissipated (that is, $u_c = B\sigma_3 = 0$), the deviator stress, $\Delta\sigma_d$, on the specimen is increased to cause shear failure.

- During the test, simultaneous measurements of $\Delta\sigma_d$ and Δu_d are made.

Consolidated – Undrained Triaxial Test

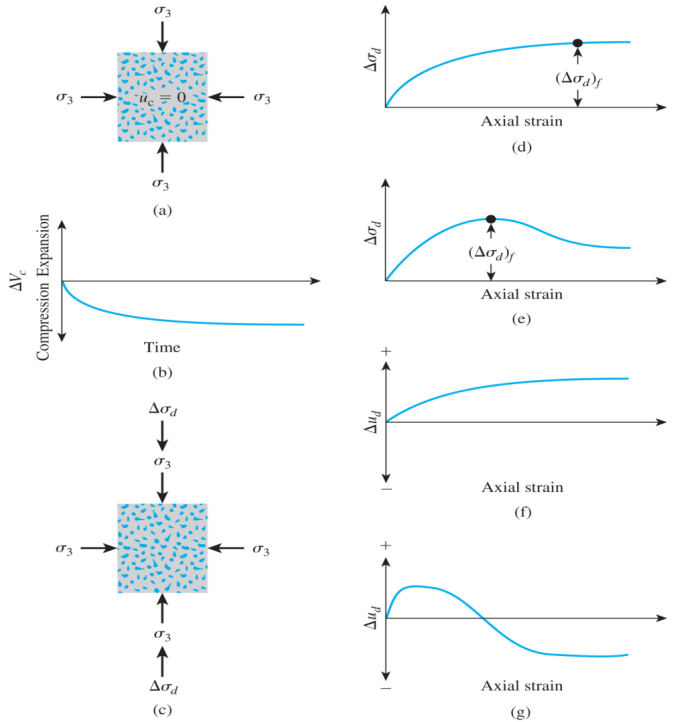


Figure 12.26 Consolidated undrained test: (a) specimen under chamber confining pressure; (b) volume change in specimen caused by confining pressure; (c) deviator stress application; (d) deviator stress against axial strain for loose sand and normally consolidated clay; (e) deviator stress against axial strain for dense sand and overconsolidated clay; (f) variation of pore water pressure with axial strain for loose sand and normally consolidated clay; (g) variation of pore water pressure with axial strain for dense sand and overconsolidated clay

Consolidated – Undrained Triaxial Test

- Pore water pressure parameter

$$\bar{A} = \frac{\Delta u_d}{\Delta \sigma_d}$$

= Skempton's pore pressure parameter (Skempton, 1954)

Consolidated – Undrained Triaxial Test

- In loose sand and normally consolidated clay, the pore water pressure increases with strain.
- In dense sand and overconsolidated clay, the pore water pressure increases with strain to a certain limit, beyond which it decreases and become negative (with respect to the atmospheric pressure).
- This decrease is because of a tendency of the soil to dilate.

Consolidated – Undrained Triaxial Test

- Major principal stress at failure (total):

$$\sigma_3 + (\Delta\sigma_d)_f = \sigma_1$$

- Major principal stress at failure (effective)

$$\sigma_1 - (\Delta u_d)_f = \sigma_1'$$

- Minor principal stress at failure (total):

$$\sigma_3$$

- Minor principal stress at failure(effective):

$$\sigma_3 - (\Delta u_d)_f = \sigma_3'$$

$$\sigma_1 - \sigma_3 = \sigma_1' - \sigma_3'$$

Consolidated – Undrained Triaxial Test

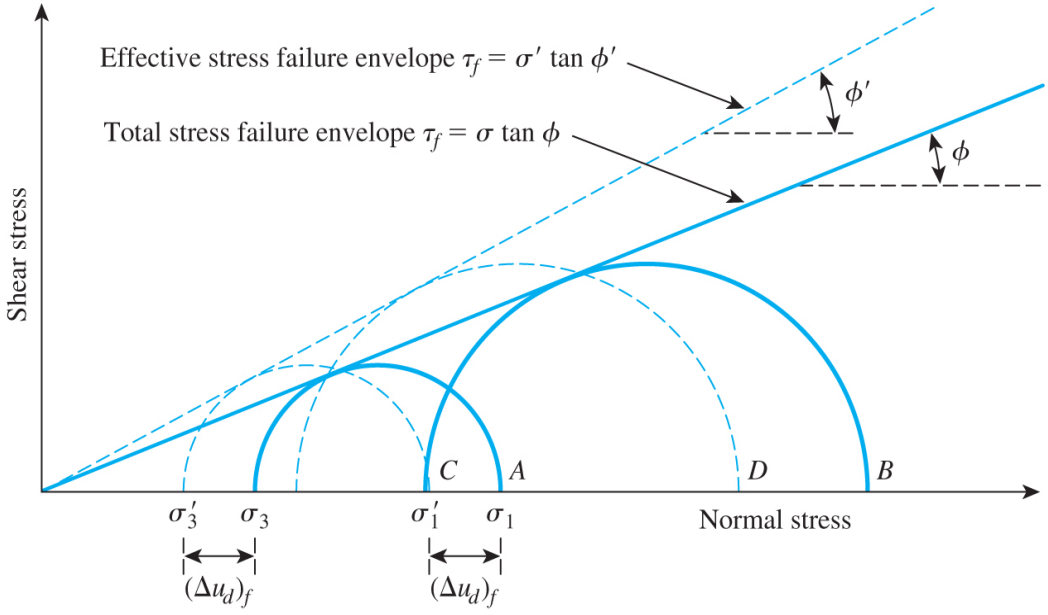


Figure 12.27 Total and effective stress failure envelopes for consolidated undrained triaxial tests. (*Note:* The figure assumes that no back pressure is applied.)

Consolidated – Undrained Triaxial Test

- For sand and NC clay

$$\tau_f = \sigma' \tan\phi$$

- For OC clay

$$\tau_f = c + \sigma \tan\phi_1$$

- Consolidated-drained tests on clay soils take considerable time. For this reason, consolidated – undrained tests can be conducted on such soils with pore pressure measurements to obtain the drained shear strength parameters.

Consolidated – Undrained Triaxial Test

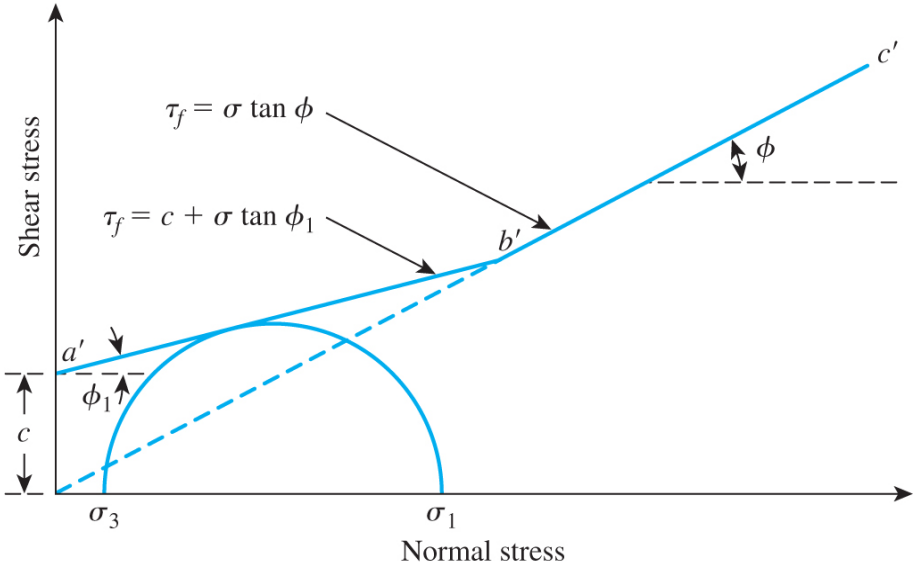


Figure 12.28 Total stress failure envelope obtained from consolidated-undrained tests in over-consolidated clay

Consolidated – Undrained Triaxial Test

- Skempton's pore water pressure parameter at failure.

$$\bar{A} = \bar{A}_f = \frac{(\Delta u_d)_f}{(\Delta \sigma_d)_f}$$

- The general range of values in most clay soils
  - Normally consolidated clays: 0.5 to 1
  - Overconsolidated clays: -0.5 to 0

Consolidated – Undrained Triaxial Test

Table 12.3 Triaxial Test Results for Some Normally Consolidated Clays Obtained by the Norwegian Geotechnical Institute*

Location	Liquid limit	Plastic limit	Liquidity index	Sensitivity[a]	Drained friction angle, ϕ' (deg)	$\bar{A}_f$
Seven Sisters, Canada	127	35	0.28		19	0.72
Sarpborg	69	28	0.68	5	25.5	1.03
Lilla Edet, Sweden	68	30	1.32	50	26	1.10
Fredrikstad	59	22	0.58	5	28.5	0.87
Fredrikstad	57	22	0.63	6	27	1.00
Lilla Edet, Sweden	63	30	1.58	50	23	1.02
Göta River, Sweden	60	27	1.30	12	28.5	1.05
Göta River, Sweden	60	30	1.50	40	24	1.05
Oslo	48	25	0.87	4	31.5	1.00
Trondheim	36	20	0.50	2	34	0.75
Drammen	33	18	1.08	8	28	1.18

* *After Bjerrum and Simons, 1960. With permission from ASCE.*

[a] See Section 12.13 for the definition of sensitivity

Consolidated – Undrained Triaxial Test

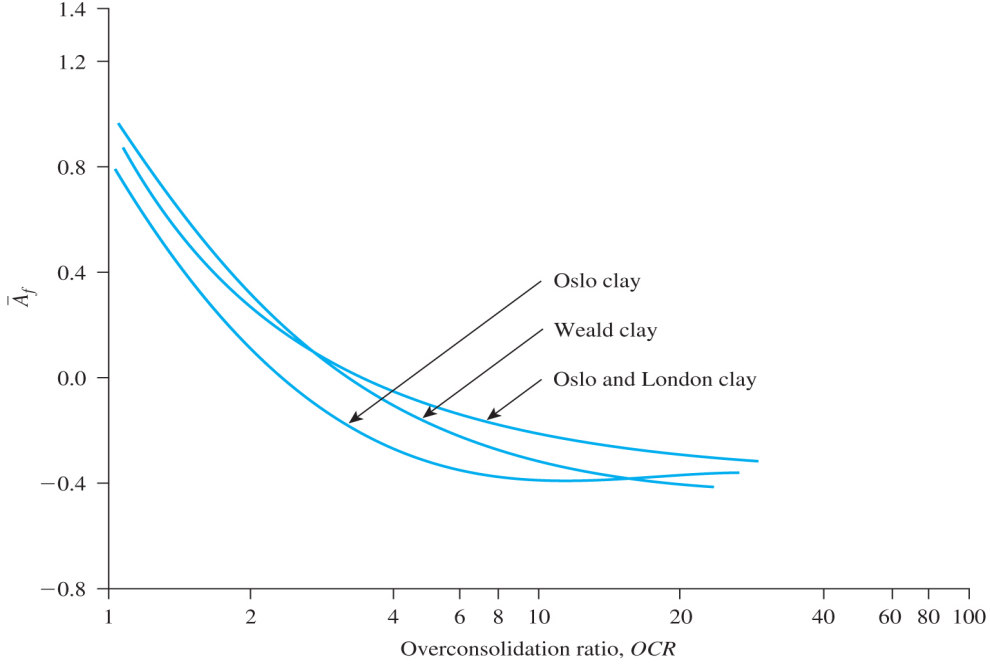


Figure 12.29 Variation of $\overline{A}_f$ with overconsolidation ratio for three clays (*Based on Simon, 1960*)

Unconsolidated – Undrained Triaxial Test

- In unconsolidated–undrained tests, drainage from the soil specimen is not permitted during the application of chamber pressure, $\Delta\sigma_3$
- The test specimen is sheared to failure by the application of deviator stress, $\Delta\sigma_d$, and drainage is prevented.

$$u = B\sigma_3 + A\Delta\sigma_d = B\sigma_3 + A(\sigma_1 - \sigma_3)$$

- This test is usually conducted on clay specimens and depends on a very important strength concept for saturated cohesive soils.

Unconsolidated – Undrained Triaxial Test

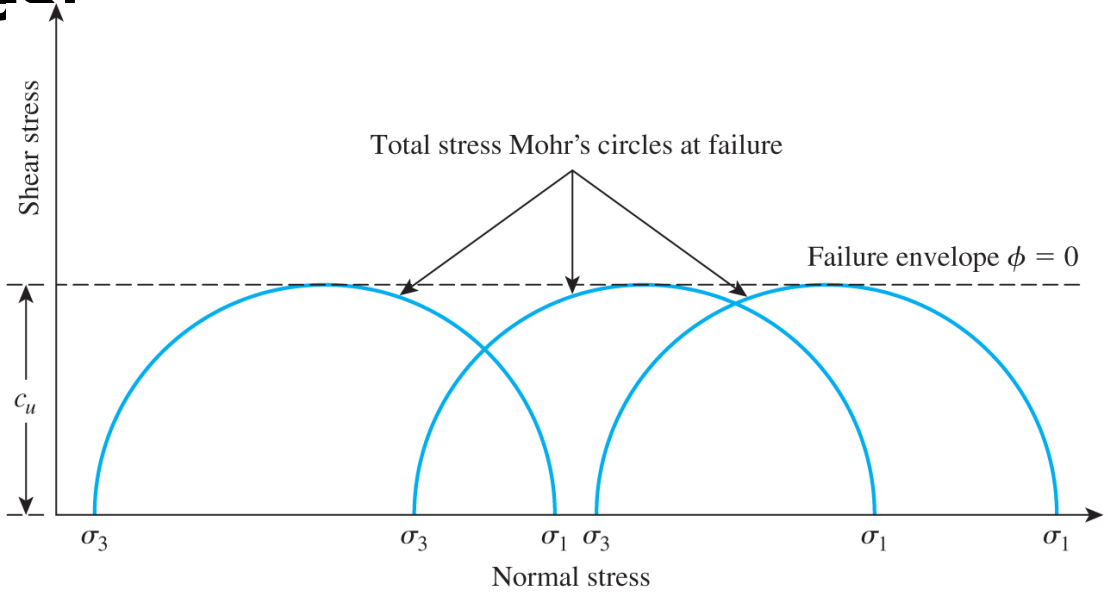


Figure 12.31 Total stress Mohr's circles and failure envelope ($\phi = 0$) obtained from unconsolidated-undrained triaxial tests on fully saturated cohesive soil

Unconsolidated – Undrained Triaxial Test

- Φ = 0 condition,
- Explanation of Φ = 0 concept
  - Specimen no.1

consolidated at and then sheared to failure without drainage

Total Mohr's circle : P

Pore pressure at failure =

- At Failure

$$\sigma_1' = [\sigma_3 + (\Delta\sigma_d)_f] - \Delta u_d = \sigma_1 - \Delta u_d$$

$$\sigma_3' = \sigma_3 - \Delta u_d$$

- Effective Mohr's circle: Q

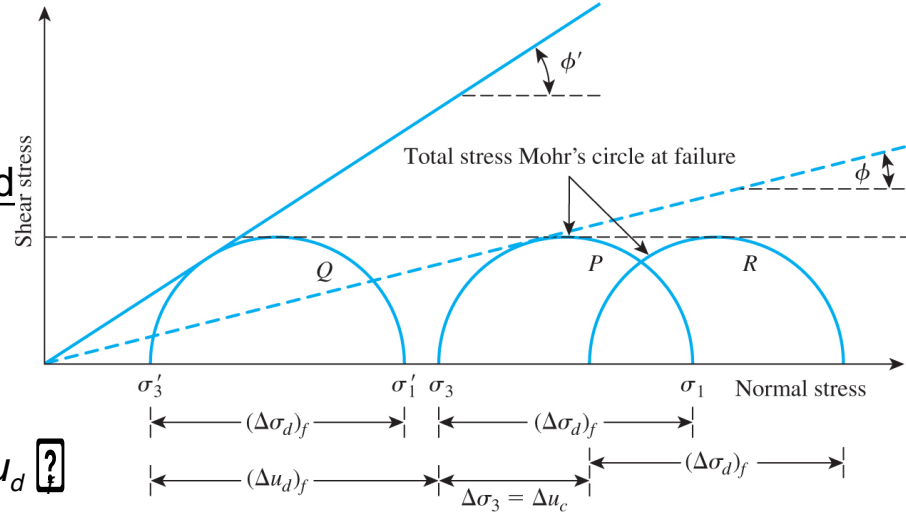


Figure 12.32 The $\phi = 0$ concept

Unconsolidated – Undrained Triaxial Test

- Specimen no.2

consolidated to σ_3 and increase by $\Delta\sigma_3$ without drainage

Total Mohr's circle: R

$\Delta u_c = \Delta\sigma_3$

Effective Confining Pressure:

$$\sigma_3' = \sigma_3 + \Delta\sigma_3 - \Delta u_c$$

$$= \sigma_3 + \Delta\sigma_3 - \Delta\sigma_3$$

$$= \sigma_3$$

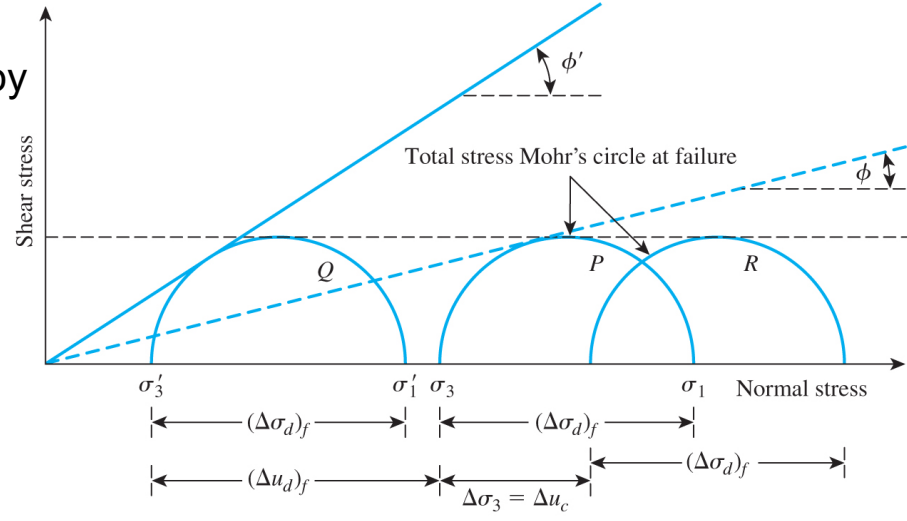


Figure 12.32 The $\phi = 0$ concept

Unconsolidated – Undrained Triaxial Test

- Specimen no.2

Then Shear → should fail at the same deviator stress $(\Delta\sigma_d)_f$ and pore water pressure $(\Delta u_d)_f$

- At failure $[\sigma_3 + \Delta\sigma_3] - [\Delta u_c + (\Delta u_d)_f] = \sigma_3 - \Delta u_{d_f} = \sigma'_3$
- And the major principal effective stress:

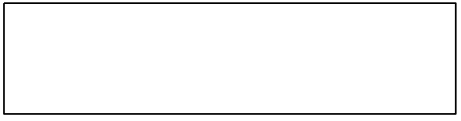
$$[\sigma_3 + \Delta\sigma_3 + (\Delta\sigma_d)_f] - [\Delta u_c + (\Delta u_d)_f] = [\sigma_3 + (\Delta\sigma_d)_f] - \Delta u_{d_f}$$

$$= \sigma_1 - (\Delta u_d)_f$$

The effective stress Mohr's circle: Q

Unconfined Compression Test on Saturated Clay

- A special type of unconsolidated-undrained test that is commonly used for clay specimens.



- q_u = unconfined compression strength

Table 12.4 General Relationship of Consistency and Unconfined Compression Strength of Clays

Consistency	q_u kN/m²
Very soft	0–25
Soft	25–50
Medium	50–100
Stiff	100–200
Very stiff	200–400
Hard	>400

Unconfined Compression Test on Saturated Clay

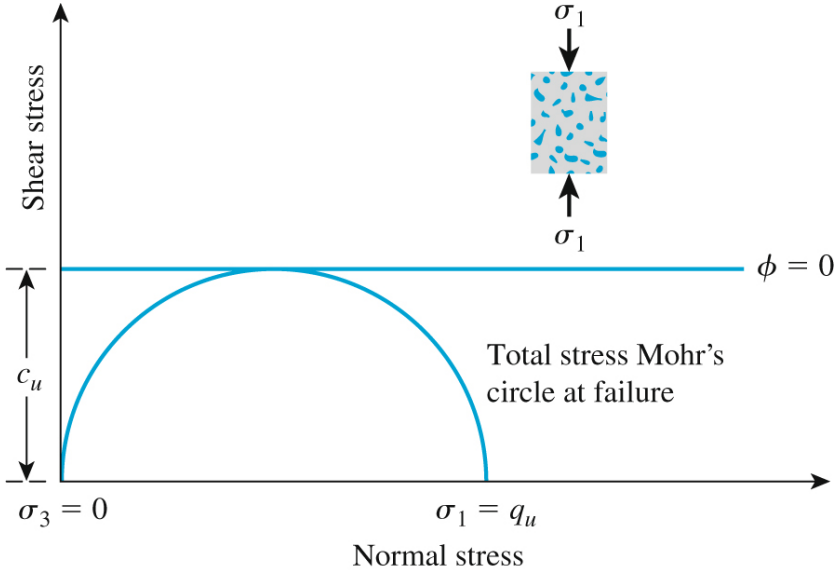


Figure 12.33 Unconfined compression test

Unconfined Compression Test on Saturated Clay



Figure 12.34 Unconfined compression test equipment (*Courtesy of ELE International*)



Figure 12.35 Failure by shear of an unconfined compression test specimen (*Courtesy of Braja M. Das, Henderson, Nevada*)

Unconfined Compression Test on Saturated Clay



Figure 12.36
Failure by bulging of an unconfined compression test specimen (*Courtesy of Braja M. Das, Henderson, Nevada*)

Shear Strength of Unsaturated Cohesive Soils

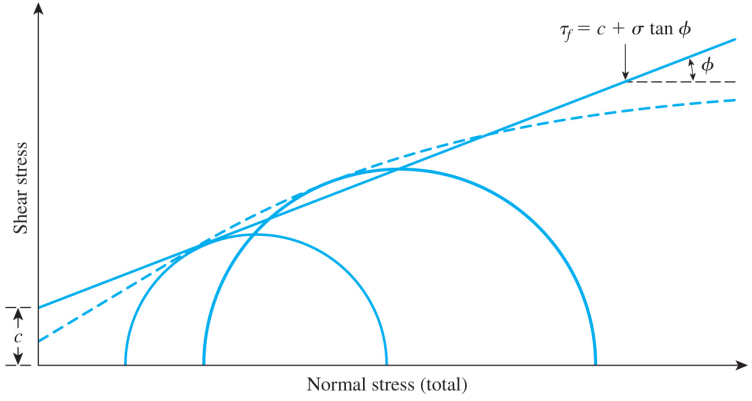


Figure 12.51 Total stress failure envelope for unsaturated cohesive soils

Shear Strength of Unsaturated Cohesive Soils

- Figure 12.52 shows the variation of the total stress envelopes with change of the initial degree of saturation obtained from undrained tests on an in organic clay.

- For design purpose the unsaturated soil specimens collected from the field must be saturated in the laboratory and the undrained strength determined.

  → Rainfall Effect

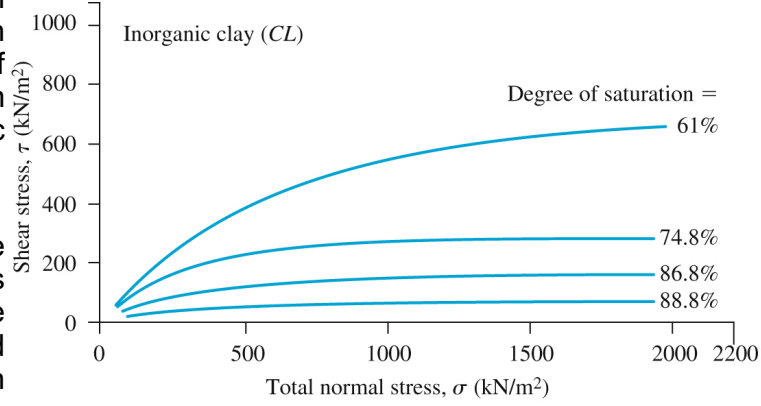


Figure 12.52 Variation of the total stress failure envelope with change of initial degree of saturation obtained from undrained tests of an inorganic clay (*After Casagrande and Hirschfeld, 1960. With permission from ASCE.*)